

Cognitive Neuroscience Research Internship (CNRI): Fall 2025



Class Time:
Tues. / Thurs.
1:25-2:40pm

Classroom Information:
Levine Science Research
Center (LSRC) B240

Contact Information:
General Communications:
cnri@duke.edu

Clubhouse Information:
LSRC
B203

Python Co-Instructors:

Joyce Park (she/her)

joyce.park@duke.edu

Office hours: T: 2:40 – 3:40

Jaime Ali Rios (he/him)

jaime.rios@duke.edu

Office hours: T 12:30 – 1:25

Conceptual Instructors:

Rotation of CCN graduate
students and postdocs. Check the
calendar!

Research Team Mentors:

Paris Brown, Robert Driver, Shanika Phillips Fullwood, Julia Smith,
Zichen “Cleo” He, Xinhao Wang

Program Overview

The CNRI is a semester-long program that introduces the basics of conducting research about human brains and behavior. *Cognitive neuroscience* is the study of how our brains allow us to think, feel, and act (including memory, attention, emotion, decision-making, and language). Our research topics and methods share elements with both psychology and neuroscience.

Interns do not need prior background in research; instead, this program will provide opportunities for learning about research methods and applications, through both career development activities and hands-on experiences. Over the semester, students will:

- gain a basic understanding of cognitive neuroscience topics.
- learn how to digest scientific articles.
- receive an introduction to coding programs relevant for cognitive neuroscience experiments (e.g., PsychoPy).
- become part of a supportive community with a track record of preparing students for success in their future research endeavors at Duke and beyond.

Course Description

Our program is structured to maximize interaction through engaging lectures, group activities/problem solving, and discussions about how to take scientific questions and translate them into behavioral tasks. Throughout the semester, you will participate in two lines of activities: **cohort-based research education** and **smaller team-based research experience**.

Cohort-Based Research Education

In our **cohort-based research education**, we strive to create a birds-eye view of life as a researcher by holding (1) conceptual classes about common approaches used to investigate human behavior and (2) Python classes to introduce the basics of computer programming.

Conceptual Classes

Conceptual classes will cover four themes:

1. Scientific Integrity, covering ethics, responsible conduct of research, and evaluating scientific claims; **2. Research Methods**, featuring mentor talks and demonstrations of diverse research approaches; **3. Beyond CNRI**, exploring professional development, graduate school and other scientific careers; and **4. Scientific Communication**, focusing on presenting research clearly, culminating in the end-of-semester poster

Python Classes

Alongside the conceptual component of the program, you will be introduced to programming in Python, with the goal of applying your knowledge of behavioral research and programming to build an experimental paradigm. During these classes, we will introduce and practice different concepts in Python and computer programming. Classes will include some lecture components but mostly focus on actively reading, writing, and editing code. Please bring a

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session.

laptop to these classes.

Team-Based Research Experience

Our team-based research experiences aim to provide a more granular look into the daily life of research. You will be divided into small teams and paired with 1-2 graduate students or postdocs. In these research teams, you will complete more specific and topical research-based activities to give an in-depth view into the every-day tasks of life as a researcher. This also gives you an opportunity to start experiencing the work expectations for research assistants (ranging from experimental design to data collection to data analysis). Our small teams also provide one-on-one mentorship to help develop research questions and interests.

You will be assigned research teams the second week of the program based on your interests and the available research projects. While we will try our best to match your interests to what is available, we will be limited in the number of spaces available on each team.

We expect research teams to meet weekly to discuss progress on tasks and projects. Each research team will differ in the topics they cover (e.g., curiosity vs. emotion regulation) and in which stage of research they are in (e.g., experimental design, data collection, data analysis), so what you do on a week-to-week basis might change as projects move along.

Core Values

Research comes with a unique set of challenges and opportunities. We encourage you to remain open to error messages (we promise there's useful info in there!) and to learn from each other (research is more fun with friends). Please reflect on our shared values that have helped us navigate these waters as we teach and collaborate with you this semester:



Course Activities & Deliverables

Class Attendance

Synchronous participation in all sessions is required. Attendance will be tracked by completing a quick Qualtrics survey during the last 5-minutes of class. After your second absence, we will require you to have a check-in meeting with leadership. After your third absence, you may lose the ability to participate in research activities.

Slack Communication

All class discussions, assigned readings, and announcements will take be posted to our Slack channel (CNRI-Intern). We request that you [download](#) Slack on both your laptop and phone, and that you enable notifications.

Coffee Chats

During the end-of-class attendance surveys, you will have the option to indicate whether you would like to meet individually with the speaker to learn more about their background or research. Your coffee and a snack will be paid for by CNRI. These chats are not required, but you are **highly encouraged** to take advantage of at least one coffee chat during the semester.

Participate in Interactive Discussions

We will be covering a lot of material in this program, most of which will be new to you all. The concepts and the nitty gritty of coding can be overwhelming and frustrating. To help ensure your learning, we

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expect active participation throughout lectures and course assignments.

1. Before each conceptual lecture, you may be asked on Slack to answer a few questions to facilitate group discussion of the journal article. Your responses to the speaker are **due on Slack the evening before the presentation**.
2. During most Python classes, there will also be weekly group homework assignments in which you apply concepts from class to complex coding problems. These problems are intended to be challenging, and we encourage you all to work on them together.
3. You should spend at least 2 hours of your weekly CNRI time commitment to being physically present in your research team's laboratory space or the CNRI Clubhouse. This can include time spent on CNRI-related assignments. Your fulfillment of this requirement will be tracked by your mentors or by a sign-in sheet in the Clubhouse.

Mid-Semester Research Oral Presentations

For the mid-semester presentation, you and the other intern on your team will prepare a set of slides to introduce the research topic you are investigating and summarize any relevant background literature (10 min. + Q&A). Example slides are available on [Box](#).

End-of-Semester Poster Presentations

You and the other intern on your team will present a conference-style research poster on the last day of CNRI. You will briefly introduce your research question and project goals, present an overview of the work you completed, and share any preliminary results and personal reflections. CNRI will pay for the [poster printing](#) at PhD Posters, which must be uploaded to **no later than Tuesday, November 25th** (pick-up on Monday December 1st).

Python Final Project

Throughout the semester, we will practice asking behavioral research questions and discussing how to translate them into code to build an experimental task. The best way to get comfortable working in Python is to practice. Therefore, you will complete (ungraded) homework components to complement the material learned in class. The fundamental skills you learn will culminate in building your own behavioral experiments in small teams.

Towards the middle of the semester, we will start to brainstorm your scientific questions of interest. This will require some background literature reviews to see what research has been done and how we can fill the gaps. You will work with your team to write a proposal that documents your literature search, scientific question, and how you plan to test your question.

At the end of the semester, the Center for Computational Thinking will meet with you to create videos about your accomplishments. These videos are intended to promote your professional development and help you learn to explain the skills you've developed from this program.

Exit Interviews

After completing the internship, we ask that you complete a brief exit interview. These interviews are analogous to a course evaluation and will gauge your understanding of the material, satisfaction with CNRI, future plans, and suggestions to improve the program.

Policies

- Recordings or other distribution of class materials are not permitted and will be considered a violation of the [Duke Community Standard](#).
- All slides and other materials from the class sessions will be posted to our **Box Folder** ([fall-2025](#)) within 24 hours after each session.

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- **Photography:** Photos may be taken for advertisement and promotion purposes. If you would prefer not to be photographed, please email us ASAP (cnri@duke.edu).
- **Email etiquette:** For all Python-related communications, please email *both* instructors (Joyce **and** Miles). For all other course-related communications, email cnri@duke.edu.
- **Payment:** You will be compensated for all CNRI-related activities (including class time, research tasks, and homework) at a rate of \$17/hour. You will be added to Duke payroll during the first class and will track your hours according to these [instructions](#).

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Wk.	Dates	Tuesday	Thursday	Due Before Class
1	09/09 - 09/11	NO CLASS	Introduction to CNRI <i>All Mentors</i>	Read Syllabus and come with any questions!
2	09/16 - 09/18	Introduction to Python	1. <u>Scientific Integrity</u> Ethics, AI, and RCR Tyler & Jade	- Complete Research Team Survey - Python HW0
3	09/23 - 09/25	Control Flow	1. <u>Scientific Integrity</u> Reading and Evaluating Literature Paris Brown	- Submit list of ideas for Python project - Python HW1 - Read article
4	09/30 - 10/02	Functions	2. <u>Research Methods</u> Cleo He	- Get feedback on projects ideas from Joyce + Jaime - Python HW 2 - Read article
5	10/07 - 10/09	Classes, Libraries, & Documentation	2. <u>Research Methods</u> Shanika Phillips Fullwood	- Work on task flow of experiment - Python HW 3
6	10/14 - 10/16	NO CLASS Duke University Fall Break	2. <u>Research Methods</u> Julia Smith	- Submit task flow of experiment - Python HW 4
7	10/21 - 10/23	Introduction to PsychoPy: Drawing to Screen	Mid-Semester Research Presentations	
8	10/28- 10/30	Introduction to PsychoPy: Saving data	2. <u>Research Methods</u> Robert Driver	- Work on python projects - Python HW 5 - Read article
9	11/04 - 11/06	Introduction to PsychoPy: Collecting Responses	4. <u>Scientific Communication</u> Presenting Research Effectively Cleo, Xinhao, Robert	- Finalize Python experiment script - Python HW 6 - Read article
10	11/11 - 11/13	Psychopy review and data visualization	3. <u>Beyond CNRI</u> Future Paths in Research Jade, Tyler, Anjou	Finalize python data collection
11	11/18 - 11/20	Python Project Experiment-a-thon!	3. <u>Beyond CNRI</u> CVs, Letters, and Networking Paris, Julia	Working version of python project
12	11/25 - 11/27	Final Presentation of Python Projects	NO CLASS: THANKSGIVING BREAK	Finalize poster presentations
13	12/02 - 12/04	Final Poster Presentations! 📄 <i>Class held from 1:00-3:00pm in LSRC B wing Atrium</i>	NO CLASS	Complete Exit Interview (sent via email)